
Symmetries in Physics — Exercise Sheet 4

Irreducibility, averaging, and Schur's lemma

Variant: Questions only

Exercise 1. Let $C_3 = \langle r \rangle$ act on $\mathbb{C}^2$ by $A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$.

- (i) Check $A^3 = 1$, and show that A is not unitary for the standard inner product.
- (ii) Average the standard inner product over the group, and check that the result M satisfies $A^\dagger M A = M$.
- (iii) A is the matrix $B((1\ 2\ 3))$ of Lecture 4's worked example. Compare M with the Gram matrix of $f_1 = e_1 - e_2$ and $f_2 = e_2 - e_3$ in $\mathbb{C}^3$, and explain the result without computing.
- (iv) Find a basis, orthonormal for M , in which A is the rotation matrix $R(2\pi/3)$.

Exercise 2. Maschke's theorem has two hypotheses. This exercise shows what each is for.

- (i) Take the representation $D(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ of the infinite group $(\mathbb{Z}, +)$ from Lecture 4's example "Reducible but indecomposable". Which step of the averaging proof cannot be carried out? Prove that no inner product on $\mathbb{C}^2$ makes ρ unitary.
- (ii) Now let the numbers be *bits*: vectors in $V = \{0, 1\}^2$, added and scaled modulo 2. Let $C_2 = \{e, a\}$ act on V by swapping coordinates. Show that the line spanned by $(1, 1)$ is invariant but has no invariant complement, and say where the averaging proof breaks here.

Exercise 3. Lecture 4's worked example ends with a remark: the standard representation of S_3 and the defining representation of D_3 are equivalent. Prove it. Here is everything you will need.

- *The standard representation* (Lecture 4). A permutation σ acts on $\mathbb{C}^3$ by $\rho(\sigma)e_k = e_{\sigma(k)}$. The subspace $V_0 = \{\sum_i v_i = 0\}$ is spanned by $f_1 = e_1 - e_2$ and $f_2 = e_2 - e_3$, and in the basis (f_1, f_2) the matrix of the cycle is

$$B((1\ 2\ 3)) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

- *The defining representation* (Lecture 3). The generators r (the rotation through $2\pi/3$) and s (the reflection in the y -axis) are represented by

$$D(r) = \frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

and the vertices of the triangle are $p_1 = (0, 1)$, $p_2 = (-\frac{\sqrt{3}}{2}, -\frac{1}{2})$ and $p_3 = (\frac{\sqrt{3}}{2}, -\frac{1}{2})$.

- *The isomorphism $D_3 \cong S_3$* (Lecture 1): $r \mapsto (1\ 2\ 3)$, $s \mapsto (2\ 3)$. It assigns to each element the permutation by which that element permutes the vertex labels.
- (i) Compute $B((2\ 3))$.
- (ii) Find an invertible T with $T B((1\ 2\ 3)) = D(r) T$ and $T B((2\ 3)) = D(s) T$. By Sheet 3, Exercise 4, checking the two generators is enough.

- (iii) Use Schur's lemma to show that every T satisfying both equations of (ii) is a multiple of the one you found.

Hint: send e_k to the vertex p_k of the triangle.

Exercise 4. Let ρ be a complex irreducible representation of a finite group G on V .

- (i) Use Schur's lemma to show that every central element z acts as a scalar, $\rho(z) = \lambda(z)\mathbb{1}$.
- (ii) Suppose moreover that ρ is faithful. Show that $z \mapsto \lambda(z)$ is an injective homomorphism from the centre $Z(G)$ to $\mathbb{C} \setminus \{0\}$, and deduce that $Z(G)$ is cyclic.
- (iii) Conclude that the Klein four-group has no faithful irreducible representation.
- (iv) Show that the defining representation of D_4 , with $D(r) = R(\pi/2)$ and $D(s) = \text{diag}(1, -1)$ as on Sheet 1, is irreducible. Check part (ii) on it: the centre of D_4 is $\{e, r^2\}$.

Exercise 5. The quaternion group is $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$.

- (i) Find all one-dimensional representations.
- (ii) Realise a two-dimensional representation with Pauli matrices, sending i, j, k to $-i\sigma_1, -i\sigma_2$ and $-i\sigma_3$.
- (iii) Verify its irreducibility by a commutant argument. Show that only scalars commute with all the representing matrices. Explain (using Maschke's theorem) why a reducible representation would admit a non-scalar commuting operator.
- (iv) The representation of (ii) is faithful. Check Exercise 4(ii) on it.

Exercise 6 (the coordinate functions carry the defining representation). This exercise works through Example 3.9 of Lecture 3, the functions x and y . Let D_3 act on the plane by the defining representation of Lecture 3,

$$D(r) = \frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

with the vertices of the triangle at $\vec{p}_1 = (0, 1)$, $\vec{p}_2 = (-\sqrt{3}/2, -1/2)$, $\vec{p}_3 = (\sqrt{3}/2, -1/2)$. Functions on the plane transform by $(\rho(g)\psi)(\vec{v}) = \psi(g^{-1}\vec{v})$.

- (i) Check that r carries $\vec{p}_1 \mapsto \vec{p}_2 \mapsto \vec{p}_3 \mapsto \vec{p}_1$. Write down r^{-1} and compute the four functions $\rho(r)x, \rho(r)y, \rho(s)x, \rho(s)y$ as linear combinations of the coordinate functions x and y .
- (ii) Write the matrices of $\rho(r)$ and $\rho(s)$ in the basis (x, y) , using the convention that the j -th column lists the components of the image of the j -th basis vector. Compare them with $D(r)$ and $D(s)$.
- (iii) Evaluate $\rho(r)x$ at the vertex $\vec{p}_1$. Which vertex does the value belong to, and why is that what the inverse in the definition demands?
- (iv) The $2p_x$ and $2p_y$ orbitals of an atom are $x f(|\vec{r}|)$ and $y f(|\vec{r}|)$, with one radial function f . How do they transform under D_3 ?